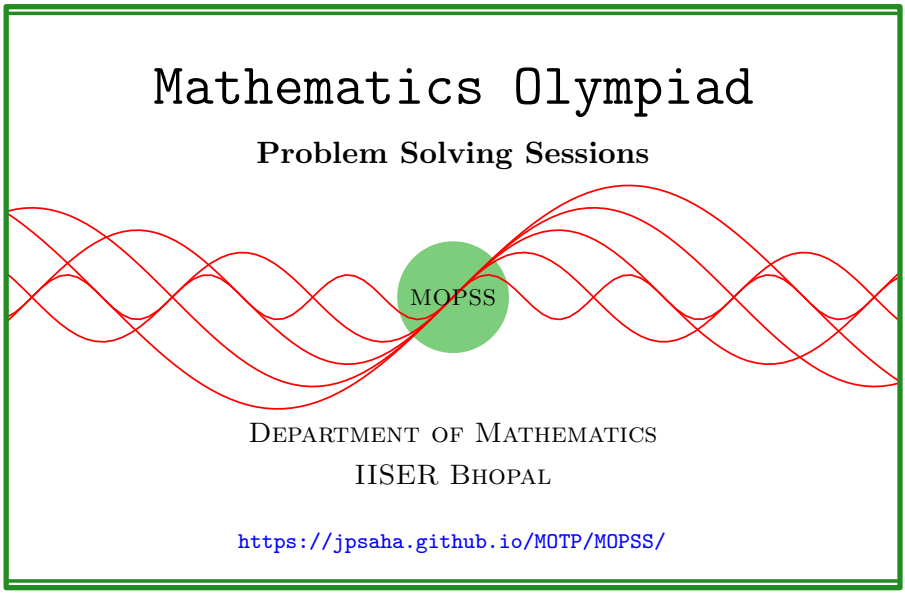
Polygons



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (IOQM 2023 P6,) 2

§1 Polygons

Exercise 1 (IOQM 2023 P6,). Let X be the set of all even positive integers n such that the measure of the angle subtended by a side at the center of some regular polygon is n degrees. Find the number of elements in X.

Summary — Determine the positive integers $r \geq 3$ such that dividing 360 by r yields an even number.

Walkthrough —

(a) Let $r \geq 3$ be an integer and consider a regular r-gon. Note that $360/r$ is even if and only if r is equal to one of

$$3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 24, 36, 45, 60, 120, 360.$$

In other words, there are precisely 16 positive integers $r \geq 3$ such that dividing 360 by r yields an even number.

(b) It follows that the number of elements of X is equal to 16.

Solution 1. Note that the map

$$r \mapsto \frac{360}{r}$$

is a bijection from the set of positive integers $r \geq 3$ such that $360/r$ is even to the set X, consisting of the even positive integers n such that the measure of the angle subtended by a side at the center of some regular polygon is n degrees. For a positive integer r, note that $360/r$ is even if and only if r divides 180. Using the prime power factorization

$$180 = 2^2 \cdot 3^2 \cdot 5,$$

we see that the number of positive divisors of 180 is

$$(2+1)(2+1)(1+1) = 18.$$

Two of these divisors are 1 and 2, which are less than 3. Thus, the number of positive integers $r \geq 3$ such that $360/r$ is even is

$$18 - 2 = 16.$$

It follows that the number of elements in X is 16. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)